

Robust Brain MRI Classification on Latent Representations Leveraging Denoising Diffusion Probabilistic Models

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Abstract

This project arises from the need to explore new techniques to improve image classification in the medical field, with particular focus on the reliability and robustness of predictions. The main motivation stems from the frequent presence of noise in medical images, which can undermine the accuracy of traditional classification models. Inspired by the recent research work "Robust Classification via a Single Diffusion Model Classification", we decided to implement a denoising diffusion model aimed at enhancing the model's ability to handle noisy data, as well as leveraging this approach for data augmentation to effectively expand the training dataset. During development, we realized that using an autoencoder to work in a latent space could simplify the process and help improve model robustness. Indeed, the latent dimension compresses information into a more compact, potentially more stable, and noise-resilient representation. The project structure involved building an autoencoder, followed by training a MLP classifier on the latent dimension output by the encoder. Subsequently, we integrated a diffusion model capable of providing noise-related information to the system, thus training the model to classify accurately even under uncertainty and image degradation. Finally, the complete model combines the encoder, a denoising module that "corrupts" and then "cleans" the latent representation, and a ResNet for final classification. This approach aims to enhance the safety and accuracy of classification in the critical medical context, demonstrating the potential of diffusion techniques and latent representations in handling noisy images.

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Introduction

Brain tumors are among the most complex and delicate pathologies to diagnose in the clinical field. In particular, early and accurate diagnosis is crucial to increase the effectiveness of treatments and improve the patient’s quality of life. However, distinguishing between healthy brains and tumor formations through visual interpretation of magnetic resonance imaging (MRI) can be extremely challenging, especially in cases where structural differences between tissues are minimal or not clearly visible. Manual analysis by expert radiologists, although essential, is subject to subjective variability, time-consuming, and not always easily scalable.

In recent years, the application of artificial intelligence, and deep learning in particular, has shown great potential in supporting physicians during the diagnostic process. Convolutional Neural Networks (CNNs) are especially effective in the automatic classification of medical images. Alongside them, more advanced techniques such as autoencoders and generative models (e.g., diffusion models) allow for reducing data complexity and improving the quality of learned representations.

In this project, we focused on the automatic classification of brain magnetic resonance imaging (MRI) scans, with the goal of distinguishing between healthy brains and patients with tumors. Specifically, we initially chose to simplify the problem by binarizing the dataset, reducing it to two classes: no tumor and meningioma. This decision was driven by computational constraints, as implementing and training complex models requires significant computing power, which is not always available in local environments. Nonetheless, the entire architecture was designed to be extendable to the 4-class scenario without requiring structural modifications.

To address the challenge of computational complexity and make the pipeline more efficient, we introduced an autoencoder to encode the images into a more compact latent space, preserving relevant information while reducing background noise. This transformation simplifies the input data for subsequent models and potentially improves the generalization capability of the classifier network. The process thus involves an initial encoding through the encoder, followed by a separate Multi-Layer Perceptron (MLP) that performs classification based on the latent features.

We then trained the denoising diffusion model to output the average

noise present in a latent dimension, given the timestep of the image being analyzed. This model is subsequently used for classification using both latent information and timestep t , making the classification noise-aware. This approach is particularly useful in real-world scenarios where images or latent representations may be corrupted, degraded, or incomplete, such as in the following cases:

- MRI images affected by artifacts or acquisition noise
- Latent representations compromised by generative or compressive processes
- Situations where classification is needed even before the denoising process is complete

Furthermore, the model is also used to denoise the latent representation itself, a functionality that is integrated into the subsequent pipeline stages. The second part of the pipeline involved the use of a ResNet, employed as an alternative to direct classification: it cleans the image via the denoiser and then classifies the filtered result.

Overall, the project combines discriminative and generative models, leveraging noise not only as a disruptive factor but also as a tool for data augmentation and robustness. The experiments conducted show that hybrid pipelines, built around encoder, diffusion model, and CNN components, can represent a promising alternative in medical classification tasks—even under limited computational resources.

The report is organized according to the logical flow of the project:

1. Dataset: description of the classes, folder structure, and preprocessing steps
2. Autoencoder
3. Comparison between Baseline Models and Autoencoder results: simple CNN on the binary task and an autoencoder architecture and classification using MLP on encoded images
4. Diffusion Model: key concepts and classification noise-aware
5. Pipeline with ResNet: denoising + final classification

6. Conclusions and Future Work: general reflections and improvement perspectives

This structure is intended to provide a comprehensive overview of the work carried out, highlighting how classic deep learning techniques and new ones can concretely support the automatic diagnosis of brain tumors through magnetic resonance imaging.

1 Dataset

The dataset used for training and evaluating the classification model consists of brain Magnetic Resonance Images (MRIs), divided into four classes:

- **notumor**: brain images without tumor presence (2000 images)
- **glioma**: presence of glioma-type tumor (1621 images)
- **meningioma**: presence of meningioma-type tumor (1775 images)
- **pituitary**: presence of pituitary-type tumor (1757 images)

The dataset is well-balanced across the different classes, with a slight predominance of the **notumor** class.

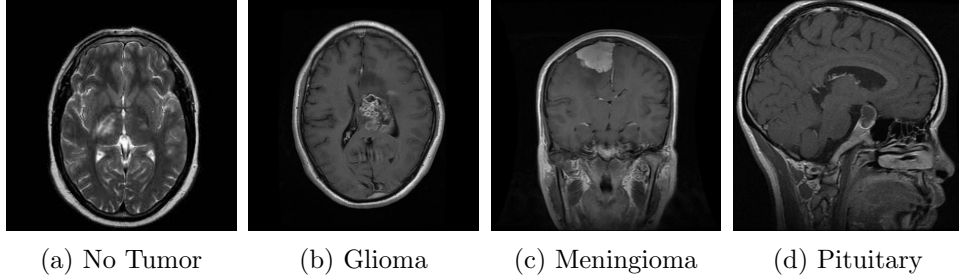


Figure 1: One MRI scan for each class

1.1 Preprocessing Steps

Before training the model, the images were preprocessed through the following steps:

- **Resizing**: all images were standardized to a size of 256x256 pixels
- **Tensor conversion**: the images were converted into PyTorch tensors
- **Normalization**: normalization was applied to each RGB channel, using a mean of $[0.5, 0.5, 0.5]$ and a standard deviation of $[0.5, 0.5, 0.5]$, bringing the pixel values into the range $[-1, 1]$

These steps ensure consistency and improve the stability of the model training process.

1.2 Labeling Strategy

The main focus of the work was on binary classification between images with and without tumors, using a targeted and simplified setup:

- **Main Phase: Binarization (notumor vs meningioma)**

In the main phase of the analysis, classification was reduced to two classes:

- **notumor**: images without tumors
- **tumor**: images containing only meningioma-type tumors

This choice allowed us to simplify the problem and focus on recognizing the presence of a specific tumor type compared to a healthy brain.

- **Final Analysis: Multiclass Classification (4 classes)**

In a later and secondary phase, an analysis was conducted on the complete original dataset, aiming to distinguish between all four classes: *notumor*, *glioma*, *meningioma* and *pituitary*. This extension allowed for the evaluation of the model's potential in multiclass classification tasks.

1.3 Data split

The dataset was split as follows:

- Separation into training and test sets
 - 80% for training
 - 20% for test
- Within the training set, a further split was performed:
 - 80% for training subset
 - 20% for validation subset
- Instead, the test set was split as follows:
 - 60% for test subset
 - 40% for calibration subset

The split was kept **stratified**, taking advantage of the dataset's folder structure organized by class, in order to preserve class proportions across the various subsets.

2 Autoencoder for Compression and Preprocessing

Autoencoders are self-supervised deep learning models designed to learn a compact representation (embedding) of data, reducing its dimensionality while preserving essential features. They are based on a mirror-symmetric architecture composed of two main components:

- The **encoder**, which transforms the original image into a low-dimensional latent vector
- The **decoder**, which attempts to reconstruct the original image from this compressed representation

2.1 Theoretical Foundations

The training objective is to minimize the difference between the original image and the reconstructed image, measured by the **Mean Squared Error (MSE)**. Through this process, the autoencoder learns to ignore redundancies, irrelevant details, or noise, and instead focuses on key structural information. In the context of this project, the autoencoder serves a dual purpose:

1. **Visual compression:** the input MRI images, which originally occupy a large space (e.g., $256 \times 256 \times 3$), are compressed into a latent vector of size 128. This transformation simplifies and lightens the processing required by subsequent models.
2. **Noise filtering and simplification:** the encoder learns a representation that tends to filter out noise, retaining only information that is meaningful for classification. This is particularly useful in clinical environments, where data may be affected by acquisition artifacts, contrast variations, or systemic noise.

The main advantage of this approach is that the subsequent network (MLP) does not need to operate on complex images, but on already-organized, salient features. This simplifies the classification task and reduces computational requirements.

Moreover the autoencoder does not require labels during the pretraining phase.

To improve the model’s trustworthiness and transparency —especially in a clinical context where decisions must be explainable— two interpretability techniques were implemented:

- **SHAP** helped assess the importance of each component in the latent vector for the final decision. The analysis revealed that some dimensions of the embedding were significantly more influential than others in classification, suggesting that the model had learned strongly discriminative latent features. Additionally, SHAP explanations were consistent across multiple images, reinforcing the system’s credibility.
- **Conformal Prediction** was implemented to provide a confidence interval for each prediction. This made it possible to formally quantify the model’s uncertainty: images predicted with high probability also tended to have high conformity levels, while ambiguous cases (near the threshold) were rightly flagged with low confidence. In practice, the system can *“know when it doesn’t know”* — a crucial capability in healthcare.

2.2 Practical Implementation

To compress the brain MRI images and extract compact, yet informative representations, a **convolutional autoencoder** was trained. The input images, preprocessed to 256×256 with three normalized RGB channels, were processed by a network composed of an encoder and a decoder.

The **encoder** uses three blocks of 2D convolutions with a 4×4 kernel, stride 2, and padding 1, followed by **BatchNorm2d** and **ReLU activations**, progressively reducing spatial dimensions and producing a latent embedding of 128 dimensions. The **decoder**, symmetrically, uses transposed convolutions (deconvolutions) to reconstruct the images from the latent vector, minimizing **MSELoss** between the original and reconstructed images.

Training the model led to visually coherent reconstructions, indicating that the most relevant information had been effectively encoded into the latent space. This value was thus selected as the final threshold for classification during testing.

The training of the autoencoder demonstrated stable and consistent behavior. The reconstruction loss (MSE) initially decreased sharply and then leveled off at very low values, with a final value of **0.0099**. This indicates that the model achieved a good ability to reconstruct the original images from their latent representations. After several epochs without significant improvements (i.e., improvements below the threshold of 0.001), the **early stopping** mechanism was triggered, automatically halting training at the **93rd epoch**, thereby preventing overfitting and optimizing computation time.

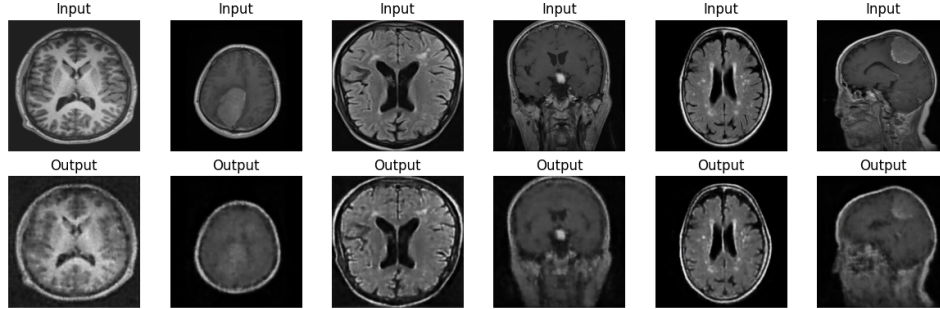


Figure 2: Images Reconstructed by the Autoencoder

2.3 Classification and Threshold Analysis

Next, the latent vector generated by the encoder was used as input for a binary classification task: distinguishing between tumor and healthy images. To this end, a model called **TumorClassifier** was built, including the pre-trained encoder and a fully connected neural network (MLP) composed of several linear layers, **BatchNorm1d**, **ReLU activations**, and **Dropout**. The model was trained using **BCEWithLogitsLoss** (Binary Cross-Entropy with Logit Loss), optimizing the distinction between classes in a probabilistic domain. The output is a probability between 0 and 1, obtained via a sigmoid function, and used to classify the image as "tumor" or "no tumor." The training performed with threshold on validation set to 0.5, achieved strong validation metrics, with:

- Accuracy above **92%**
- Well-balanced **precision** and **recall**: an **F1 score** $> 93\%$ indicates a good tradeoff between sensitivity and specificity

To binarize the model's output (i.e., decide the cutoff point beyond which the predicted probability indicates tumor presence), the system's behavior was analyzed over a wide range of thresholds from **0.1 to 1.0**, with steps of **0.01**.

For each threshold value, key metrics were computed: **accuracy**, **precision**, **recall** and **F1 score**. In medical contexts, the priority is to **avoid false negatives**, meaning not to miss detecting a tumor. Therefore, **recall** was chosen as the key metric, even at the cost of increasing false positives.

From the analysis, a threshold of **0.33** was found to offer the best tradeoff:

- Higher Accuracy (93.27%)
- Very high **recall** (98.21%)
- Stable F1 score (93.99%)
- Acceptable **precision** (90.12%)

This value was thus selected as the final threshold for classification during testing.

The use of **SHAP (SHapley Additive exPlanations)** enabled a detailed analysis of which dimensions of the latent vector (128-feature embedding) most strongly influence the binary classification between tumor and non-tumor images. Specifically, the method assigns a (positive or negative) contribution to each feature in relation to the model’s predicted probability, offering a mathematically grounded criterion for interpretation.

Numerically, it emerged that only a subset of the embedding components are consistently relevant for the decision: the model tends to rely on a **small number of latent dimensions**. This suggests that the encoder was able to compress the discriminative information into a few key features, increasing the system’s **efficiency and interpretability**.

Visually, the heatmaps generated by SHAP — where **red areas indicate a positive impact** on the “tumor” classification and **blue areas a negative impact** — show both interesting and somewhat problematic patterns:

- The SHAP maps highlight **strong activation along the skull edges**, especially in regions with high contrast between the white bone and the black background. This behavior suggests the model tends to focus on **generic structural signals**, likely learned during the compression phase by the autoencoder. However, such focus is **not always clinically relevant** and could introduce decision noise, causing the model to rely on **repetitive but non-pathological patterns**.
- On the other hand, in the presence of clear tumors, the maps also show **red hotspots** precisely in the **tumor regions**, i.e., in areas of high contrast with surrounding tissue. This is a **positive signal**, as it demonstrates that the model can indeed detect visual anomalies consistent with tumor masses.

In summary, the model exhibits a dual sensitivity:

- To **high-contrast zones** (white on black), which often include tumors but also normal structures like the skull
- To **localized anomalous areas**, where a well-defined mass is present

This observation is consistent with the encoder’s architecture, which likely learned to compress intensity gradients as distinguishing elements. However, it also reveals a **potential attention bias** toward **generic visual features**.



(a) SHAP for a positive MRI



(b) SHAP for a negative MRI

Figure 3: Examples of SHAP

A **Conformal Prediction** system was implemented to evaluate and quantify the uncertainty of predictions generated by the classifier based on the combination of the encoder and a fully connected neural network. This technique - particularly valuable in medical contexts — not only enables binary classification (*tumor vs no tumor*), but also allows each prediction to be associated with a **formal confidence level**, helping to understand how strongly the model “believes” in its output.

To achieve this, a split conformal approach was used. This involves dividing the dataset into a separate calibration subset, distinct from the actual test set. The method computes a threshold parameter called $\hat{q}$ based on the calibration scores. This value determines which predictions are reliable enough

to be considered with high confidence. In our case, with a significance level $\alpha = 0.1$, the goal was to guarantee an expected coverage of at least 90% - meaning that at least 90% of the true labels would be included in the prediction sets generated by the model.

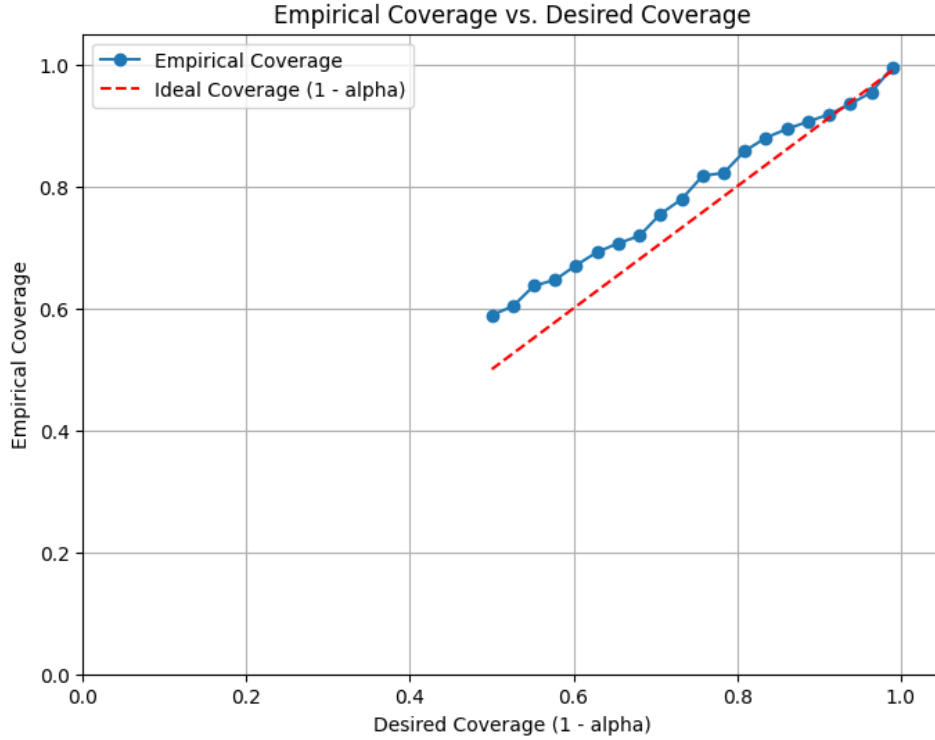


Figure 4: Empirical Coverage vs Desired Cover for Autoencoder + MLP

The results were very positive. The empirical coverage reached 91.35%, exceeding the desired threshold. This means that in over 91% of cases, the correct class was included in the model’s prediction set, indicating reliable and consistent performance. The computed value of $\hat{q}$ — the discriminative threshold for accepting a prediction with confidence — was 0.4438. In other words, whenever the predicted probability exceeded this value, the model considered the prediction sufficiently confident.

Another important aspect is the average size of the prediction set, which was 0.9808. This indicates that in the vast majority of cases, the model was able to return a single class as the prediction, confirming its ability to

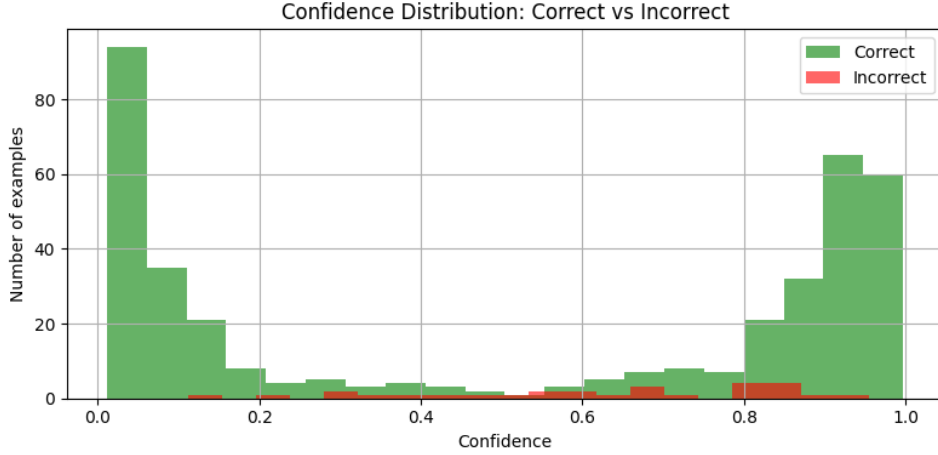


Figure 5: Confidence Distribution for Autoencoder + MLP

make clear and decisive classifications. The bar chart of the “Distribution of Prediction Set Sizes” visually confirms this behavior, showing that nearly all predictions resulted in a set of size one, with only a few rare cases returning an empty set.

Finally, the “Empirical Coverage vs. Desired Coverage” plot shows a trend very close to the theoretical ideal line $1 - \alpha$, indicating that the system is well-calibrated across the full range of required confidence levels.

In summary, Conformal Prediction demonstrated that our classifier is not only capable of producing accurate predictions, but also of attaching a reliable measure of uncertainty to each one.

The observed behavior suggests that the system is able, in a sense, to “know when it doesn’t know”.

The chart shows the distribution of confidence scores for correct (green) and incorrect (red) predictions made by the model composed of the encoder and fully connected network. This type of visualization is extremely useful for assessing the calibration of the model—that is, how well the predicted probability (confidence) reflects the actual likelihood of the prediction being correct.

From the chart, it is clear that the model tends to be very confident when it is correct: most of the correct predictions are concentrated near extreme confidence values, particularly around 0.0 and 1.0. This is likely due to the binary nature of the task and the distribution of predicted probabilities after applying the chosen threshold. In contrast, incorrect predictions (in

red) are more evenly distributed, but tend to cluster in the central regions, especially between 0.4 and 0.6, where the model is less confident. This evidence aligns with expected behavior: low-confidence predictions are more often associated with errors, while high-confidence predictions (close to 1 or 0) are generally correct. In particular, it is notable that incorrect predictions rarely reach extreme confidence values, suggesting that when the model is wrong, it tends to be less certain.

3 Comparisons for validation

3.1 Binary classification (Encoder + MLP vs CNN)

In this study, a comparison was made between two binary classification approaches applied to the same dataset. On one hand, a traditional convolutional neural network (CNN) was used, receiving the raw images as input. On the other, a model was adopted that includes a **preliminary compression phase via our autoencoder**, followed by classification in the latent space using a MLP. The goal was to assess whether the compressed representation produced by the encoder could maintain — or even improve — the predictive quality compared to a more standard approach.

3.1.1 Quantitative Results

The numerical results reveal some interesting differences:

- The **CNN** achieved:
 - **Accuracy**: 92.62%
 - **Precision**: 98.28%
 - **Recall**: 85.73%
 - **F1 score**: 91.56%
- The **Encoder + MLP** approach achieved:
 - **Accuracy**: 93.27%
 - **Precision**: 90.12%
 - **Recall**: 98.21%
 - **F1 score**: 93.99%

This comparison shows that the **embedding-based approach** demonstrates higher sensitivity, as evidenced by its significantly higher recall.

On the other hand, the higher precision of the CNN indicates a **more conservative behavior**: when it predicts a tumor, it’s rarely wrong—but it may miss real cases (i.e., more false negatives).

This tradeoff is reflected in the **F1 score**, which balances precision and recall: the **encoder-based model** performs better overall, suggesting that the latent representation preserves—and possibly enhances — the **discriminative features** needed for classification.

3.2 Binary Classification (Encoder + ResNet vs ResNet)

Following these positive results, the same experimental structure was repeated using a more sophisticated network — **ResNet** — to compare the performance of **embedding + ResNet** with that of a ResNet applied directly to the original data. This allowed for a deeper exploration of the potential of latent embeddings, even in more complex modeling contexts.

This model performed very well on the test data, getting a value close to 97.4% for each of the main classification metrics, implying an effective separation between MRIs containing a meningioma and those representing a healthy brain. Nevertheless, keeping in mind that we’re working in the medical field, which is extremely sensitive, a high overall accuracy, precision, or F1-score might not be sufficient: a model obtaining a better recall could be superior since it minimizes the number of false negatives, this hints that previous Encoder + MLP model could be more suitable than this Encoder + ResNet.

Now we compare this model to another ResNet architecture that works directly on the images: we took a ResNet50 architecture with weights pre-trained on the ImageNet dataset, removed its classifier head and frozen all its layers, then we added a new classification head and trained only this new part. In this way we obtained great results with all the main classification metrics being close to 98%. While these results are excellent, the marginal performance gain over previous models does not fully justify the significantly increased complexity of this larger architecture.

However, the good performance of these models further supports image classification by working on latent spaces.

3.3 Comparative Evaluation on the 4-Class Classification Task

A comparative analysis was conducted between two distinct approaches applied to the 4-class classification task on brain MRI images. The first approach is based on a pretrained encoder followed by a fully connected network, while the second adopts a CNN built from scratch and trained directly on the original images. Both models used the same multiclass dataset (including images of glioma, meningioma, pituitary tumor, and healthy subjects), ensuring fair conditions for performance evaluation.

In the first case, the encoder previously developed and trained on the binary classification task (no tumor vs meningioma) was reused with its structure left unchanged, and re-trained on the full multiclass dataset. The encoder compresses MRI images—originally sized $256 \times 256 \times 3$ — into a 128 - dimen-

sional latent vector, preserving the most salient information while filtering out noise and redundant components. A fully connected network composed of multiple dense layers with BatchNorm, ReLU, and Dropout was then applied to this latent representation to perform the final classification across the four categories.

In the second approach, a simple yet complete convolutional network was built: three Conv2D + MaxPooling blocks, followed by a Flatten layer and a fully connected classifier with Dropout. The input to this network consists of the original images, without passing through a compression or latent pre-processing phase. The model was trained using categorical cross-entropy and the Adam optimizer, and evaluated on the test set via `.evaluate()`.

The comparison revealed a significant performance gap. The encoder + fully connected model achieved around 96% accuracy, with high and well-balanced macro metrics (precision, recall, and F1-score), demonstrating its ability to effectively discriminate between all classes—including those that are typically more difficult to distinguish, such as glioma and meningioma. The CNN, on the other hand, reached a final accuracy of approximately 89.3%, exhibiting less stable behavior and greater difficulty in generalizing. It’s important to emphasize that both architectures were evaluated on the same images and under the same conditions, ensuring a fully fair comparison.

3.3.1 Interpretation and Implications

From an interpretative standpoint, these results suggest that information compression via encoder not only reduces the dimensionality of the problem—making it more efficient from a computational perspective — but also contributes to greater predictive accuracy. The encoder acts as an intelligent filter, selecting the most relevant features for discrimination and removing non-informative variations or systematic noise. This significantly simplifies the task for the classifier, which no longer has to extract raw features from high-resolution images, but instead operates on an already-optimized embedding.

In summary, the encoder + fully connected network approach proved superior to the CNN, both in terms of absolute accuracy and its ability to maintain high performance across all classes. In a clinical setting, where classification reliability is crucial and data can be highly heterogeneous, this strategy offers a clear advantage in terms of robustness, scalability, and interpretability.

4 Diffusion-Based Classification on Latent Representations

In this section of the project, we explore the use of a Denoising Diffusion Probabilistic Model (DDPM) as an alternative tool for MRI image classification — operating not directly on the images, but on their latent representations obtained via an encoder. The core idea is to leverage the generative power of the diffusion model for a discriminative task, utilizing its denoising mechanism to assess whether a latent vector belongs to a specific class.

The process begins with the latent vector generated by the autoencoder’s encoder. Instead of classifying this embedding directly, Gaussian noise is injected into it following the typical diffusion schedule — a forward process where noise is gradually added. The neural network is trained to predict components (e.g., noise or clean data) from noisy embeddings, but the full reverse diffusion process is not performed during training.

In this work, we do not train a full generative DDPM model. Building upon the idea of adding noise to latent representations, we first developed a probabilistic denoiser that learns to estimate the noise distribution within the latent vector. We then extended this approach by designing a noise-aware classifier that predicts class labels while simultaneously leveraging noise estimation as an auxiliary signal. This enables improved classification robustness on noisy latent inputs without explicitly reconstructing the clean data, effectively using noise information to guide classification.

4.1 Denoising Diffusion Probabilistic Model

4.1.1 Theoretical Description of the LatentDenoiserResidual Model

The `LatentDenoiserResidual` model is a neural network designed to perform noise estimation in the latent space of encoded data, inspired by the principles of diffusion probabilistic models. Its goal is to predict the noise distribution—specifically, the mean and variance parameters—that have been injected into a latent vector during a forward diffusion process.

Input and Time Embedding The model receives as input a noisy latent vector $z_t \in \mathbb{R}^d$, where d is the dimension of the latent space, corrupted by Gaussian noise at diffusion timestep t . To incorporate temporal information about the noise level, the model uses a sinusoidal time embedding function, which maps the scalar timestep t into a high-dimensional representation

$\mathbf{e}_t \in \mathbb{R}^d$. This embedding enables the network to condition its predictions on the current noise level, which is critical for denoising tasks.

The noisy latent vector z_t and the time embedding $\mathbf{e}_t$ are concatenated, resulting in a vector of dimension $2d$, which is then normalized via Layer Normalization to stabilize training.

Residual Architecture The concatenated and normalized input is passed through an initial linear layer projecting it to a higher-dimensional hidden space. The model then applies two consecutive residual blocks, each comprising Layer Normalization, the SiLU (Sigmoid Linear Unit) activation, dropout for regularization, and linear layers. These residual blocks allow the network to learn complex transformations while mitigating vanishing gradient issues, encouraging stable and efficient training.

Output Parameterization After the residual blocks, the features undergo a final normalization and a linear projection to an output vector of size $2d$. This vector is split into two parts:

- The first d dimensions represent the predicted mean μ of the noise distribution.
- The second d dimensions represent the predicted log-variance $\log \sigma^2$ of the noise distribution.

To ensure numerical stability and positivity of the variance, the log-variance values are clipped and exponentiated to produce the variance σ^2 . The standard deviation σ is computed as the square root of the variance.

Model Objective By estimating both the mean and standard deviation of the noise, the model effectively learns a heteroskedastic Gaussian noise distribution conditioned on the noisy latent input and the diffusion timestep. This probabilistic output allows for more expressive modeling of uncertainty compared to a fixed variance assumption, improving the denoising capability in downstream tasks such as latent space classification or reconstruction.

4.1.2 Training Procedure and Results

The training of the probabilistic denoiser was conducted by freezing the encoder of the pretrained autoencoder to extract fixed latent representations z_0 of the input images. At each training step, a random diffusion timestep t

was sampled, and Gaussian noise was injected into the latent vector following the forward diffusion process $q(z_t | z_0, t)$. The denoiser network then took the noisy latent z_t and the timestep t as inputs and predicted both the mean and standard deviation of the noise distribution.

The loss function optimized was the **Negative Log-Likelihood (NLL)** of the Gaussian noise, computed by combining the predicted mean and variance with the actual noise added. This probabilistic loss encourages the model not only to predict the noise accurately but also to quantify uncertainty in its predictions.

Over 20 epochs, the training loss consistently decreased from approximately 0.65 to 0.46, indicating effective learning and improved noise estimation. This demonstrates the model’s capability to robustly capture the noise distribution at various diffusion timesteps, forming a solid foundation for subsequent tasks such as noise-aware classification or latent denoising.

4.2 Noisy-aware Classifier

4.2.1 Pipeline Description

To enhance the robustness of our MRI latent vector classification against noise, we evolved our approach from a pure denoising model to a noise-aware classifier. While the initial denoiser focused solely on estimating and removing noise in latent embeddings, the noise-aware classifier integrates noise information directly into the classification process. This allows the model to leverage noise characteristics as an auxiliary signal, improving its ability to predict the correct class even when inputs are degraded.

The sequence of steps includes:

- **Creation of the Noisy Dataset:** Each latent embedding is transformed into a sequence of noisy versions, each corresponding to a time step t of the diffusion process. Thus, every embedding generates multiple degraded variants, all labeled correctly (tumor or no tumor).
- **Design of the Classifier Model:** A lightweight fully connected neural network conditioned on diffusion time steps, designed to take a noisy latent vector as input and predict the probability of the ‘tumor’ class. The classifier is trained on noisy embeddings sampled across all diffusion steps.
- **Training and Validation:** Training is performed on these noisy latent vectors. The underlying assumption is that a robust classifier

should accurately predict classes even with noise, especially at early diffusion steps where noise intensity is still moderate. Performance metrics such as accuracy, precision, recall, and F1 score are tracked throughout training.

- **Evaluation and Testing:** After training, the model is tested on the noisy test set to evaluate its robustness as noise increases. The evaluation also measures up to which diffusion step t the classifier remains reliable before noise obscures the signal beyond recognition.

4.2.2 Results and Interpretation

The success of this method is supported by strong performance metrics, particularly in terms of accuracy and F1 score, even under noisy conditions. This suggests that:

- The embedding generated by the autoencoder indeed captures the most relevant information for classification.
- The diffusion-based denoising can be effectively used not only for image generation but also for reliable class recognition.

From a medical perspective, this approach could be extremely valuable in real clinical scenarios, where data quality is not always optimal—images may be noisy, incomplete, or acquired under suboptimal conditions. A model that maintains high performance on degraded inputs represents a significant advantage, as it reduces the risk of diagnostic errors when dealing with compromised images.

4.2.3 Threshold Optimization for Clinical Decision-Making

A crucial aspect is selecting the optimal decision threshold for classifying images based on the predicted probability. The code performs a sweep across thresholds — from 0.1 to 0.99 — calculating classification metrics at each level.

The final threshold selected is 0.47, as it offers a balanced trade-off between accuracy and precision, while also ensuring a very high recall of 96.41%.

At this threshold:

- Accuracy = 93.27%
- Precision = 91.49%
- F1 Score = 93.89%

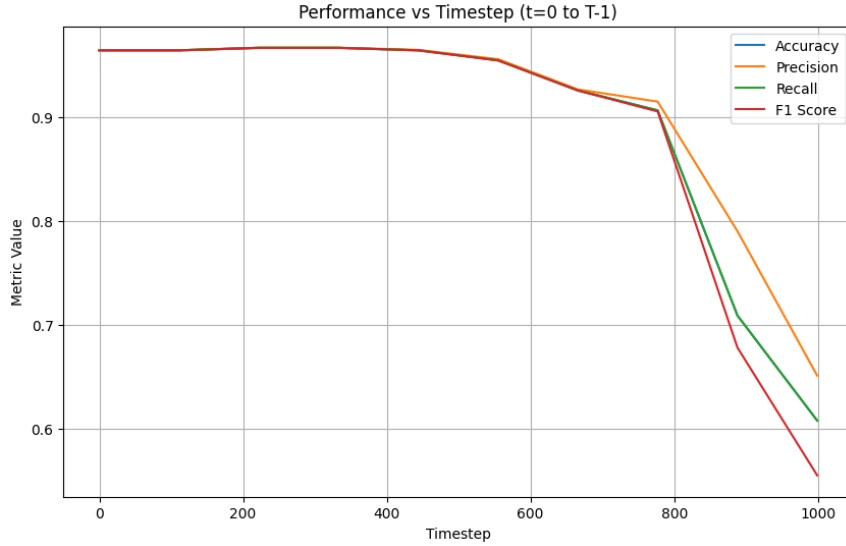


Figure 6: Performance at different timesteps, given the right t

4.2.4 Performance Evaluation Over Time

Performance is evaluated comprehensively across all key metrics—accuracy, precision, recall, and F1 score — throughout a wide range of timesteps from 0 to 999. The results show that the model maintains excellent classification performance up to approximately $t = 500$. For instance:

- At $t = 333$, the F1 score reaches 96.63%
- At $t = 444$, performance remains high, with $F1 = 96.39\%$

However, after $t = 666$, performance begins to drop sharply, and by $t = 888$, the F1 score falls to 67.83%, eventually reaching a critical low of 55.55% at $t = 999$.

This trend is clearly illustrated in the “Performance vs. Timestep” chart, where a steep decline appears after step 600. The conclusion is that the network is robust to moderate noise, but fails when the image becomes too severely degraded.

Noise-Awareness Under Misinformed Timesteps

In addition to the standard evaluation, where the classifier is provided with the correct timestep t for each noisy input, we performed a more challenging test to assess the model’s intrinsic noise-awareness capabilities. In this

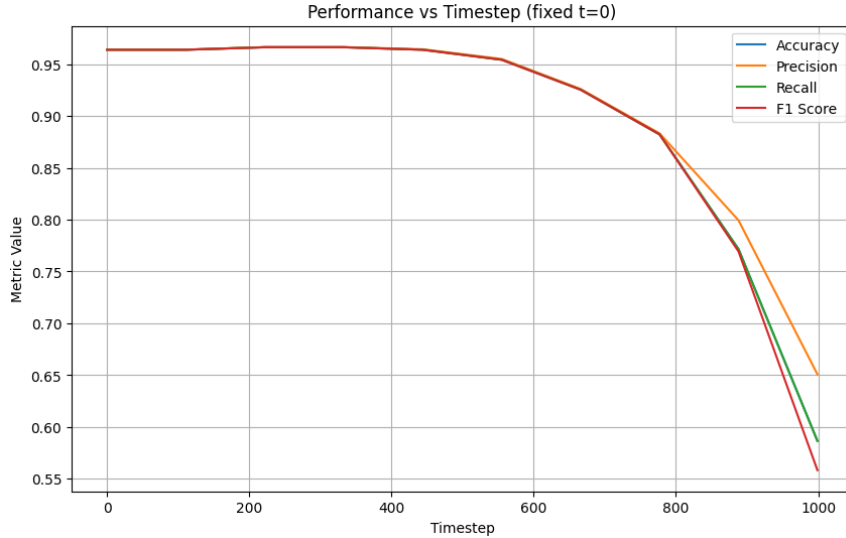


Figure 7: Performance at different timesteps with $t=0$ as input

setting, the classifier was evaluated on the same sequence of increasingly corrupted latent inputs (from $t = 0$ to $t = 999$), but with a fixed and incorrect timestep $t = 0$ supplied at all times—effectively making the model believe that every input was clean.

Surprisingly, the model continued to exhibit robust performance across a wide range of actual noise levels. It maintained F1 scores above 95% up to approximately $t = 500$, and only experienced a performance drop when the noise became extremely severe. For instance, at $t = 888$, the F1 score dropped to 77.29%, and further declined to 54.25% at $t = 999$.

These results highlight an important characteristic of the model: rather than relying solely on the timestep metadata, the classifier appears to have learned to interpret the level of corruption directly from the latent representation z_t . This suggests that it has developed an internal mechanism for estimating noise and adjusting its decision-making accordingly, even when misinformed about the actual timestep.

Such resilience is especially advantageous in real-world scenarios. The ability to perform reliable classification under such uncertain conditions supports the model’s practical utility and reinforces its suitability for deployment in sensitive applications.

4.3 Comparison Between the Denoising Diffusion Classifier and the Tumor Classifier

This section presents a comparison between the proposed Denoising Diffusion Model (DDM) classifier and the Tumor Classifier introduced earlier, which operates on the same latent representations but without explicit noise conditioning. Both models share the same pretrained encoder, however only the DDM classifier is trained to handle latent inputs corrupted at varying noise levels through conditioning on the diffusion timestep.

The comparison evaluates classification performance on inputs corrupted at different diffusion timesteps, aiming to assess the advantage provided by the noise-aware diffusion-based training in maintaining robust classification under degraded conditions.

By leveraging the denoising mechanism inherent in the diffusion process, the DDM classifier learns to extract meaningful features from latents at multiple noise intensities, thus enhancing its resilience to severe corruption. In contrast, the Tumor Classifier processes the same latent features but lacks explicit adaptation to noise or timestep information, leading to reduced performance when noise is significant.

Standard classification metrics—accuracy, precision, recall, and F1-score are used to assess performance across the diffusion timeline. The results, presented in tabular and graphical forms, demonstrate that the DDM classifier consistently outperforms the Tumor Classifier as noise levels increase, highlighting the benefits of noise-aware modeling for this clinical classification task.

4.3.1 Performance Comparison Across Timesteps

From the evaluation results, performance metrics have been computed across a range of diffusion timesteps (from 0 to 900), under identical experimental conditions for both the DDM-based classifier and the Tumor Classifier. For each selected timestep, both models receive input representations corrupted at the same noise level, ensuring a fair comparison.

The results demonstrate a consistent and substantial advantage of the DDM classifier over the Tumor Classifier across all noise levels:

- **At timestep 0** (i.e., clean inputs), the DDM achieves an accuracy of 96.39%, while the Tumor Classifier achieves 93.27%.
- **At timestep 500**, with moderate noise, the DDM maintains high performance with 96.39% accuracy, whereas the Tumor Classifier drops

sharply to 74.04%.

- **At timestep 900**, under heavy degradation, the DDM still achieves a respectable accuracy of 73.56%, while the Tumor Classifier gets outperformed at 72.36%.

This trend is consistent across all metrics: precision, recall, and F1 score. The DDM classifier exhibits remarkable robustness to noise: while the Tumor Classifier’s performance begins to deteriorate rapidly even at low noise levels (e.g., timestep 100), the DDM remains highly stable up to timestep 600. This suggests that noise conditioning during training enables the DDM classifier to extract useful semantic information even from significantly degraded latent representations.

Figures 8 and 9 illustrate this pattern: the curves associated with the DDM remain high and relatively flat across the entire diffusion timeline, while those of the Tumor Classifier decline steeply, particularly in F1 score and recall. These observations highlight the inherent noise-resilience of the DDM-based approach.

Moreover, confidence histograms provide additional insight. At timestep 0, the DDM exhibits a clear bimodal distribution with predictions clustered around 0 and 1 — indicative of high certainty. As noise increases (e.g., at timestep 111 or 777), the distribution progressively shifts toward the decision boundary (0.3–0.5), reflecting increasing uncertainty. Nonetheless, the DDM retains higher confidence than the Tumor Classifier across all noise levels.

A fixed threshold of 0.47 was used to binarize predictions from both models, ensuring consistency during evaluation.

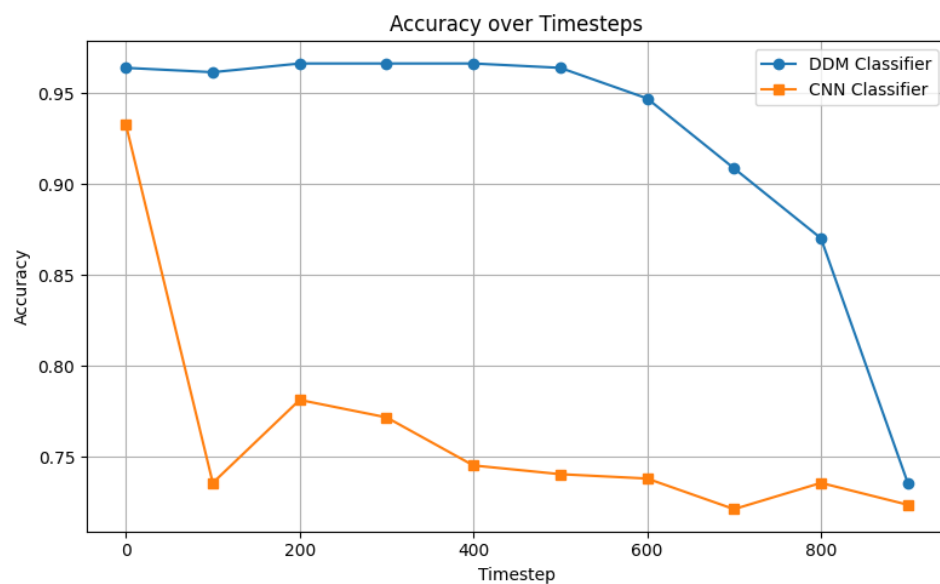


Figure 8: Accuracy for DDM and tumor classifier at each timestep (DDM misinformed)

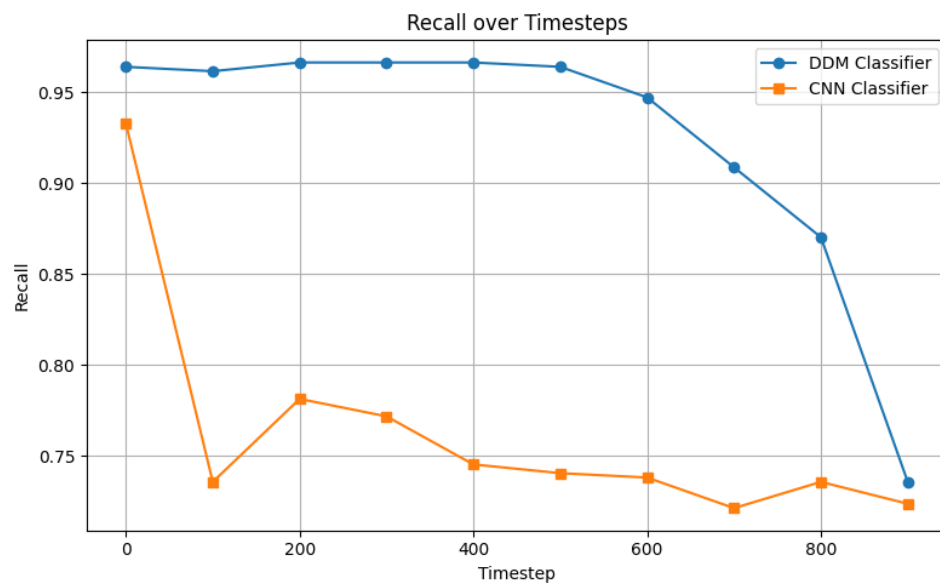


Figure 9: Recall for DDM and tumor classifier at each timestep (DDM misinformed)

5 Hybrid Pipeline: Encoder, Denoising Diffusion, and ResNet for Robust Classification

In the final phase of the project, a more advanced pipeline was implemented, composed of three components:

1. An **encoder** to compress MRI images into a latent space,
2. A **denoising diffusion model (DDM)** to clean noisy latent representations,
3. A **ResNet** to perform the final classification.

This architecture was designed to assess whether denoising the latent space could enhance classification performance, particularly under noisy conditions.

5.1 Theoretical Framework

The DDM is trained to reconstruct clean latent representations from Gaussian-corrupted inputs. This process serves as a noise-aware transformation that enhances the signal-to-noise ratio before classification.

By placing the ResNet after the DDM, the classification stage operates on purified and more informative features, enabling more accurate predictions. This separation of responsibilities — denoising first, decision-making second — is expected to improve generalization and robustness.

5.1.1 Testing and Results

The model was evaluated across multiple diffusion timesteps, each corresponding to a different level of degradation. For each timestep, standard classification metrics were computed: **accuracy**, **precision**, **recall**, and **F1-score**. Average metrics across all timesteps were also calculated to assess overall robustness.

Results confirmed that the pipeline is effective, especially under moderate noise. Using a classification threshold of 0.22 — chosen to prioritize recall in a clinical setting — the model maintained high sensitivity while preserving competitive precision.

5.2 Practical Evaluation

The complete architecture — encoder, denoiser, and ResNet — demonstrated strong classification performance. Each component contributes strategically:

- The **encoder** extracts compact latent features,
- The **DDM** denoises these features via a reverse diffusion process,
- The **ResNet** performs binary classification on the refined latent space.

With the threshold set to 0.22, the model achieved:

- **Accuracy:** 92.79%
- **Precision:** 90.72%
- **Recall:** 96.41%
- **F1 score:** 93.48%

Performance was further supported by a near-perfect **ROC curve** with an **AUC of 98.1%**, and a high-quality Precision–Recall curve, indicating strong predictive confidence even in the presence of class imbalance.

5.2.1 Interpretability and conclusions

The **confusion matrix** reveals a high rate of correct classifications, with only **8 false negatives out of 223 positive** cases — a critical strength for medical applications such as brain tumor diagnosis, where failing to detect a positive case can have serious consequences.

The model was also assessed in terms of calibration, analyzing the confidence distribution of correct vs. incorrect predictions. The chart shows that high-confidence predictions (> 0.9) are mostly **correct (green)**, while **errors (red)** are concentrated in lower-confidence areas, indicating **good model calibration**.

The **conformal prediction results** further support the reliability of the classifier:

- The **average prediction set size** was 0.9399, meaning the model typically made single-label predictions — a sign of high confidence.
- The **empirical coverage** matched the **desired coverage** precisely.

- The **coverage vs. desired coverage plot** showed points closely aligned with the ideal diagonal, confirming the **model’s reliability** from a statistical calibration perspective.

Overall, the **Encoder + DDM + ResNet** pipeline demonstrates **high effectiveness** in the task of tumor classification. It leverages:

- The **informational compression** provided by the encoder,
- The **noise resilience** of progressive denoising,
- The **discriminative power** of ResNet.

Compared to previous architectures, this model **balances strong performance** across all key metrics, statistical robustness, and interpretability reliability — making it a **promising choice** for **real-world clinical scenarios**.

6 Conclusions

This project explored various machine learning approaches for classifying MRI images to distinguish between tumor and healthy cases, primarily leveraging latent representations generated by an autoencoder. The findings highlight the effectiveness of integrating advanced techniques like latent space classification, Conformal Prediction and Denoising Diffusion Probabilistic Models (DDPMs) to create robust, interpretable and uncertainty-aware diagnostic tools.

A classifier which uses a pre-trained encoder to generate latent representation for a fully connected neural network, demonstrated strong performance with an accuracy above 93%, an high recall, larger then 98% and an F1 score above 93%. This underscores the power of latent representations in compressing relevant information for classification, leading to efficient and accurate models.

Direct comparisons revealed the significant advantage of using the autoencoder’s latent space for classification: The Encoder + MLP approach achieved a higher F1 score with respect to the CNN trained on the images (93.99% vs. 91.56%) and extremely superior recall (98.21% vs. 85.73%). While the CNN had higher precision, the encoder-based model’s ability to minimize false negatives made it the more suitable choice for the context we’re working in. This indicates that the latent representation preserves and enhances the discriminative features, simplifying the classifier’s task.

Similarly, the Encoder + ResNet model performed exceptionally well, with metrics close to 97.4%. Even more complex direct ResNet architectures, while achieving slightly higher overall metrics (near 98%), did not justify their increased complexity.

This reinforces the argument that latent spaces offer an optimal balance between performance and computational efficiency. These comparisons collectively underscore that information compression via the encoder not only reduces dimensionality but also contributes to greater predictive accuracy and robustness, making it a clear advantage for clinical applications.

The project’s exploration of Denoising Diffusion Probabilistic Models (DDPMs) for latent space classification demonstrated remarkable resilience to noisy inputs. The LatentDenoiserResidual model successfully learned to estimate noise distributions, improving its ability to classify even degraded latent vectors.

The noise-aware classifier pipeline proved highly effective, maintaining excellent classification performance up to moderate noise levels ($t=500$),

with an F1 score reaching 96.63% at $t=333$. This is a critical advantage for real-world medical scenarios where image quality can be compromised.

Moreover, the "Noise-Awareness Under Misinformed Timesteps" experiment yielded a surprising and positive outcome: the model maintained high performance (F1 scores above 95% up to $t=500$) even when told the input was clean. This suggests an intrinsic ability to interpret noise levels directly from the latent representation, making the model exceptionally resilient and practical for uncertain real-world conditions.

The comparison between the DDM classifier and the original Tumor Classifier unequivocally showed the DDM's superior robustness to increasing noise, consistently outperforming it across most noise levels. This highlights the substantial benefits of explicit noise conditioning during training for maintaining reliable performance under degraded conditions.

The final hybrid pipeline, combining the encoder, Denoising Diffusion Model (DDM) for denoising, and a ResNet for classification, achieved great overall results: by strategically placing the DDM to purify latent features before classification, the system demonstrated good accuracy (92.79%), precision (90.72%), F1 score (93.48%) and, most importantly, a high recall (96.41%).

This project developed and validated a series of advanced machine learning models for MRI image classification. By focusing on latent space representations, quantifying uncertainty, and building noise-aware classifiers: the research presents a robust, interpretable, and clinically relevant framework for brain tumor diagnosis.

References

- Angelopoulos, A.N. and Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv. doi:<https://doi.org/10.48550/arxiv.2107.07511>.
- Bishop, C.M. and Bishop, H. (2023) Deep learning: Foundations and Concepts. Springer Nature.
- Chen, H., Dong, Y., Wang, Z., Yang, X., Duan, C., Su, H. and Zhu, J. (2023). Robust Classification via a Single Diffusion Model. arXiv. doi:<https://doi.org/10.48550/arxiv.2305.15241>.
- Chen, H., Dong, Y., Shao, S., Hao, Z., Yang, X., Su, H. and Zhu, J., 2023. Your diffusion model is secretly a certifiably robust classifier. arXiv. doi:<https://doi.org/10.48550/arXiv.2402.02316>
- Luo, C (2022). Understanding Diffusion Models: A Unified Perspective. arXiv . <https://doi.org/10.48550/arXiv.2208.11970>
- Molnar, C. (2019). Interpretable machine learning : a guide for making Black Box Models interpretable. Morisville, North Carolina: Lulu.
- Rombach, R. et al. (2022) High-Resolution Image Synthesis with Latent Diffusion Models. 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10674–10685. <https://doi.org/10.1109/cvpr52688.2022.01042>.
- tom.backert (2024). Brain tumor dataset. [online] Kaggle.com. Available at: <https://www.kaggle.com/datasets/tombackert/brain-tumor-mri-data>.